

Visual Extraction of Interaction Patterns Guided by Hierarchical Clustering and Process Mining

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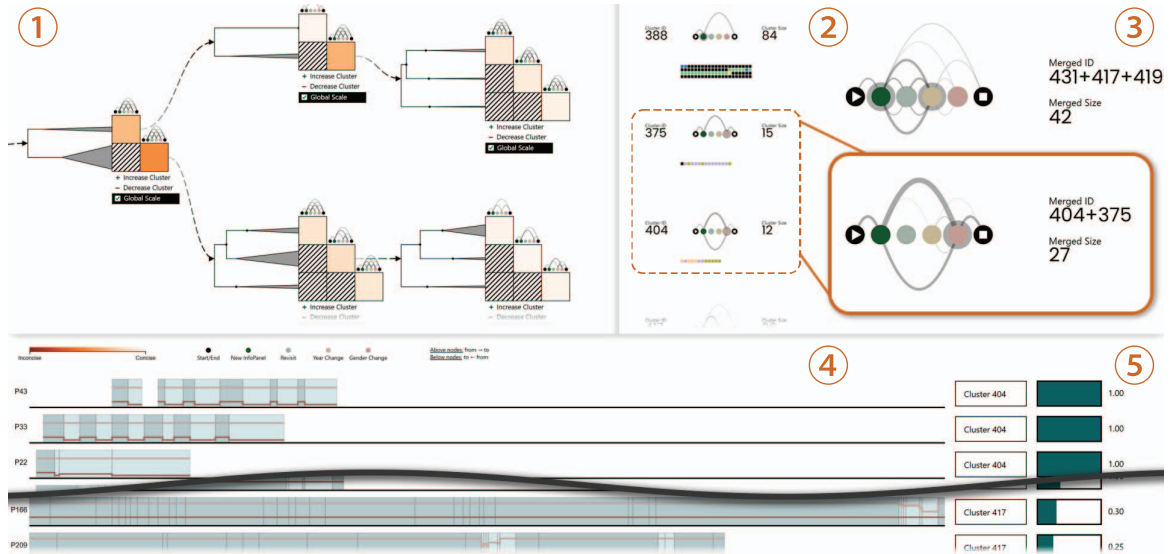


Fig. 1: The main components of reVISID. The CLUSTER AND PROCESS VIEW, featuring (1) *Cluster Pane* that summarizes the top levels of the hierarchical clustering structure using a partial dendrogram with triangular glyphs and a lower-triangular similarity matrix. (2) *Process Description Pane* presents cluster descriptions derived through process mining techniques and supports cluster description comparison. (3) Users may manually select process models for merging based on expert-guided assessments via the *Process Merging Pane*. (4) The *SEQUENCE VIEW* presents individual sequences corresponding to selected clusters, enabling detailed behavioral analysis. (5) Finally, users can evaluate how well each individual sequence conforms to a selected process model based on conformance scores.

Abstract—Understanding user interactions in digital systems is essential in analyzing user behaviors and improving system usability. However, a collection of interaction sequences is often large and unstructured, making it challenging to uncover interaction patterns. To address this challenge, we introduce a visual analytics approach that integrates hierarchical clustering and process mining techniques to support analysts in exploring unstructured, large interaction sequence data. Our system employs a tailored dynamic time warping-based similarity measure to enable comparison of interaction sequences. Based on the sequence similarities, we provide stepwise, interactive navigation of clustering results with contextual visual cues for refinement and validation. We further apply process mining to characterize derived clusters. Through these hierarchical clustering and process mining steps, analysts can progressively uncover meaningful interaction patterns while utilizing visual guidance and incorporating domain expertise. We demonstrate our system's effectiveness and applicability through two case studies involving system designers, developers, and domain experts.

Index Terms—Pattern discovery in interaction logs, visual analytics, dynamic time warping, hierarchical clustering, process mining

1 INTRODUCTION

Analyzing user interactions is essential to gain deeper insight into user behavior within interactive digital systems, playing a key role in various domains, such as system design and education [64]. Large volumes of interaction data—typically unstructured—are continuously collected to uncover patterns across many users, including common

exploration strategies and learning activities. However, deriving meaningful patterns from such data is challenging due to their complexity, heterogeneity, scalability, and interoperability [12, 13, 42, 65]. To address these challenges, various methodologies have leveraged analysis techniques such as pattern mining [12, 61, 62], process mining [69], clustering [28, 42, 66], topic modeling [67], and statistical modeling [45, 66].

Among such existing techniques, clustering is widely adopted to identify groups with similar interaction behaviors that reflect particular strategies. However, the effectiveness of clustering depends heavily on parameter choices—particularly decisions regarding the number of clusters while ensuring the quality of each cluster [3, 20, 21]. Visual analytics (VA) approaches have great potential to address this challenge by integrating domain knowledge into the exploration of retrieved clusters. VA approaches, for example, can enable analysts to interactively refine clusters through guided partitioning and merging of clusters while supporting flexible navigation within a clustering process (e.g., using a dendrogram used for hierarchical clustering).

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Another key challenge is the characterization and interpretation of retrieved clusters, which, in the context of interaction logs, corresponds to understanding how clusters relate to and reflect distinct user behaviors. This interpretation is non-trivial, as interaction sequences are often heterogeneous, noisy, and influenced by task ambiguity or user intent. Analysts must not only discern meaningful behavioral patterns from the clustered data but also assess automated clustering with domain knowledge. VA approaches integrating interactive visualization with data mining methods show substantial potential to characterize clusters of interaction sequences [12, 61, 62, 69]. Techniques for process discovery, model refinement, and conformance analysis are effective when paired with appropriate visualization and interaction methods [26, 29, 44, 60].

In our previous work [70], we designed and conducted a structured experiment with science center visitors, collecting controlled interaction logs alongside screen recordings and user questionnaires to enable a systematic analysis of user behavior. We developed a VA prototype system, named VISID, to facilitate the identification of task-solving exploration strategies. In our current work, we analyze similar interaction log data from the same exhibition setting while broadening the scope to include uncontrolled, unstructured, in-the-wild interaction data.

This extended scope introduces new analytical challenges related to data scale, variability, noise, and lack of predefined user exploration profiles. To address these, a more flexible approach that integrates robust sequence similarity measures and interactive tools to manage irregular and noisy interactions is needed. To this end, we introduce a VA approach designed to facilitate the exploration, guided navigation, and analysis of unstructured interaction data. Our approach integrates process mining with interactive visualization to support analysts in navigating large, complex clustering hierarchies and in identifying and validating user exploration strategies.

The design and development of our approach are guided by prior collaborations with domain experts, during which we systematically characterized the data and elicited domain-specific requirements. To formalize our findings, we defined a set of analytical goals and tasks, which serve as the foundation for our visualization and computational techniques. The contributions of this work are summarized as follows:

- **A novel visual analytics approach.** A VA workflow tailored for stepwise, interactive exploration and navigation of hierarchical clustering structures.
- **Process mining integration.** A process mining-driven approach for automated cluster characterization and evaluation based on process model extraction and conformance checking.
- **Tailored similarity measure.** A dynamic time warping (DTW)-based similarity measure designed to compare interaction event sequences collected in uncontrolled, real-world environments.
- **Interactive system.** We implement our approach in a VA system, named *reVISID* (*re*visiting the *VIS*ualization of Interaction *D*ynamics), aiding analysts in identifying and extracting exploration strategies in uncontrolled, in-the-wild data.

Finally, we present insights from two case studies and discuss several key aspects, including limitations and future directions.

2 RELATED WORK

We review existing work in three key areas: (1) computing and visualizing event sequence similarity; (2) summarizing event sequences via clustering and clustering representations; and (3) integrating process mining to support and enhance VA workflows.

2.1 Event Sequence Similarity Measures and Visualization

Various event sequence similarity measures have been developed to facilitate sequence comparison, each offering distinct advantages and limitations [43, 56]. These developed measures can be classified based on data type and the underlying methodological framework, such as lock-step and elastic measures [2, 19, 51, 72]. Lock-step measures, such as the Euclidean distance [22], perform a direct point-wise comparison of values between two sequences. These measures are sensitive to temporal drifts, misalignments, noise, or “distortion” in the sequences [47]. Elastic measures overcome these issues by allowing sequences to be “stretched” or “compressed” to detect similar patterns. One prominent

example is DTW [5], which aligns sequences that vary in speed or have temporal shifts, improving the robustness in assessing similarity.

In this work, we primarily examine discrete event sequences where events occur in irregular periods and event attributes may be categorical or numerical. Due to this irregularity, elastic methods such as DTW are suitable to calculate the similarity of discrete event sequences [2, 34]. User interaction logs usually consist of multiple event types (e.g., different interaction commands), forming multi-dimensional time series with irregular periods. To handle multi-dimensional time series, we can apply dependent or independent DTW methods [54]. The dependent DTW method generates a unified warping path across all dimensions simultaneously, ensuring that temporal alignment is maintained across multiple attributes. In contrast, the independent DTW method applies standard DTW to each dimension separately and aggregates the individual distance calculations to derive a final similarity measure. Our tailored similarity measure employs the independent DTW method as user interactions rarely involve multiple events simultaneously (Sec. 5.2).

In addition to similarity measures, visualization methods have been developed to compare groups of event sequences [60]. A common approach is to simplify and visualize groups of sequences by aggregating sequences based on their frequent patterns [60]. For example, ViSeq [12] represents event sequences as a graph/network by connecting events based on their transitions and applies hierarchical edge bundling to the graph to reveal common event transitions. Another example uses treemap-based visualizations to provide a hierarchical overview of groups of similar sequences [32]. Matrix-based visualizations can also highlight statistical differences in event sequences [38, 66, 67]. Similarly, we leverage a matrix-based design, integrating a heatmap and glyphs to represent aggregated sequences and facilitate sequence summarization while mitigating visual clutter.

2.2 Event Sequence Summarization

Event sequence summarization plays a critical role in grouping sequences to assist the discovery of common behaviors, anomalies, and major progression patterns. Clustering is commonly used to summarize sequences [29], with methods like *k*-means [23, 57] and hierarchical clustering [3, 4]. When focusing on interactive clustering of event sequences, hierarchical clustering provides a strong benefit as it produces a dendrogram: a tree structure that depicts the hierarchical relationships between data points and clusters [46]. Thus, the dendrogram provides information regarding clustering quality and cluster sizes at each hierarchy level, enabling the identification of outliers and undesired grouping. Our VA workflow builds upon a hierarchical clustering approach but extends it to support flexible navigation and informed dendrogram partitioning for cluster extraction. Additionally, we introduce the capability to merge similar clusters that are not adjacent in the dendrogram, enhancing customizability and adaptability in the clustering process.

Similar to our work, various clustering-based visualizations have been developed to provide a high-level abstraction and overview. For example, researchers employed treemap-based visualization to represent the hierarchical clustering outcomes of event sequences [25, 42, 63]. In contrast, Wei et al. [65] utilized a self-organizing map to locate similar click-stream clusters near each other. Other works introduce visual encodings tailored for their targeting analysis. VASABI [45] depicts one cluster as a sequence of connected circles and arranges multiple clusters in a matrix-like layout to facilitate comparative analysis. Sequence Synopsis [14] uses minimum description length for clustering and organizes clusters with color-encoded rectangle glyphs. Our work provides a *stepwise, visually guided exploration* of hierarchical clustering for event sequences. Unlike other works, our interactive system provides complementary visual cues to provide deeper insights into the clustering relationships and the underlying process models. In particular, we integrate process mining methods to enhance the interpretability of clusters by summarizing their underlying event sequences [17, 59]. One closely related work was developed for cybersecurity [16] and applies process mining to extract attack models as well as hierarchical clustering to summarize the extracted large models. Our approach differs from this: it further facilitates the identification, exploration, and validation of process models through interactive VA. Additionally,

we employ visual cues and process graphs derived from process mining methods to support informed navigation in cluster construction.

2.3 Process Mining in Visual Analytics

VA and process mining have evolved independently, with little mutual awareness and cross-disciplinary integration [26, 44], and traditionally prioritized mining methods [15, 50]. As a result, each field has developed distinct foundational concepts and terminology, limiting their convergence and collaborative advancements. Additionally, there is limited support for human-centric decision-making and analytical reasoning through interactive visualizations [18, 33]. Consequently, there has been a growing emphasis on the need for deeper integration between visual analytics and process mining [26, 37, 44] and the development of novel visualization and interaction techniques [24, 35, 49].

Until recently, limited research had explored integrated approaches leveraging concepts from both process mining and visual analytics in a cohesive manner. Among these works, the use of directed graphs to represent the process model is commonly utilized [27, 31, 41]. While these works applied graph layout algorithms to visualize directed graphs, our work employs an arc diagram design instead. This design mitigates the complexity and visual clutter in process model visualization [44].

Establishing explicit connections between mined process representations and their original data sources remains a significant open challenge [71]. A few visualizations provide summarized representations of individual event sequences to identify common sequence paths [8, 55]. To identify deviations and discrepancies from existing process models, researchers computed and visualized the conformance of individual event sequences [1, 9]. We introduce a novel workflow that integrates conformance analysis and establishes a connection between individual event sequences and the process model, facilitating analysts to identify recurring and discrepant patterns in individual sequences.

3 DOMAIN CONTEXT: THE SWEDEN IN NUMBERS EXHIBIT

The interaction log data used to exemplify our approach was collected from a digital environment called “Sweden in Numbers” (SiN). SiN is an interactive public exhibit designed to support visual exploration of official demographic and social data about Sweden and is housed at the Norrköping Visualization Center. The system displays an interactive map of Sweden, augmented with access to statistical summaries of demographics. Visitors can interact with the map by tapping on geographical regions, which triggers the appearance of bubble-shaped information panels (infopanel). These infopanel display statistics for the selected region. Within each infopanel, users can further explore characterizing attributes such as *Gender* and *Year*. Available interaction functionalities include a slider that facilitates temporal navigation across years and filter controls that enable refinement by gender. The SiN exhibit logs interactions only when visitors give their explicit informed consent through a confirmation prompt. A comprehensive description of the exhibit is provided in our previous work [70].

By analyzing users’ interactions with the exhibit, museum guides, educators, and researchers can better understand the needs, wants, and challenges of visitors. The analysis results can also inform the redesign and refinement of the exhibit to better suit the visitors’ expectations. While observational studies can also provide valuable data on these topics, they tend to be limited by their low generalizability, low scalability, and high cost. Our system uses the SiN exhibit as a case study, as its interaction logs are well-suited to exploring the needs of museum curators, guides, and educational researchers [70].

4 SYSTEM DESIGN REQUIREMENTS AND TASKS

Our design is grounded in three interconnected motivational goals: (1) To extract meaningful patterns (e.g., exploration strategies) from unstructured, in-the-wild interaction log data; (2) to flexibly identify and group interaction sequences into coherent strategies while enabling domain experts to intervene, validate, and refine clusters using contextual knowledge and analyses; and (3) to assist the identification by providing mechanisms to characterize and interpret the resulting clusters, facilitating an informed and guided interactive discovery process. We developed and refined the proposed system through an iterative

process guided by identified key design requirements. These were derived primarily from an extensive literature review, including critical analyses of features and limitations of existing approaches (described in Sec. 2), and the authors’ experience and insights obtained through close collaboration with domain experts in the analysis of interaction logs in the previous work [70]. In this section, we detail a set of four high-level design requirements (R1–4) and nine key analytical tasks (T1–9) that the system should support.

R1 - Scalable identification and exploration of sequence clusters. Real-world interaction logs, by nature, often contain large volumes of event sequences, reflecting heterogeneous user behaviors over extended periods of time. To support analysis at this scale, the system must provide a visually scalable, flexible means of abstraction and summarization, such as clustering. The data should be partitioned into meaningful clusters of distinct interaction strategies.

T1. *Summarization of clustering results.* Clustering outcomes such as dendrograms and similarity matrices can be excessively large, producing overwhelming amounts of information for analysts. To address this scalability challenge, it is essential to develop a compact yet informative representation of clustering results.

T2. *Flexible clustering detail-on-demand.* The system should provide analysts on-demand access to fine-grained information to better understand individual cluster composition. As analysts navigate deeper into the clustering hierarchy, the resulting visualizations will become spatially demanding. Thus, the system should have navigational capabilities such as zooming and panning.

R2 - Cluster interpretation. To facilitate analysis of clusters, the system should provide a descriptive summary of their constituent event sequences. This can be achieved by generating process models capturing the dominant patterns within each cluster. Further, the system should help analysts contextualize summarized patterns, evaluate each cluster’s consistency, and retrace the origin and variation of patterns.

T3. *Algorithmic assistance of cluster descriptions.* The system should support automatic generation of cluster descriptions with algorithmic approaches, such as process mining techniques. However, to accommodate domain-specific insights, analysts should be able to manually adjust related parameters, such as the number of clusters.

T4. *Event-sequence integration.* The system should establish a connection between abstracted process representations and their corresponding individual event sequences. This supports granular analysis of behavioral patterns and the identification of representative, conforming, or outlier sequences within each cluster.

R3 - Visual comparison. The system should facilitate effective comparative analysis of clusters by incorporating intuitive visual cues that enable users to quickly assess and compare cluster characteristics (e.g., conciseness or similarity).

T5. *Cluster comparison.* The system should support the visualization and comparison of clusters, both across and within clusters, i.e., inter- and intra-cluster comparisons [21]. Essential properties of each cluster, such as its size, conciseness, or the distance between clusters, should be explicitly represented.

T6. *Cluster description comparison.* In parallel with cluster comparison, the system should support effective visualization and comparison of cluster descriptions. Process mining-based descriptions can be complex and visually cluttered [44]. The system should, thus, utilize concise encoding strategies to represent cluster descriptions.

R4 - Human in the analysis loop. In addition to incorporating algorithmic support, the system must be able to incorporate domain expertise throughout the analytical workflow to enable knowledge-based validation and interactive refinement of computational results.

T7. *Human validation of cluster descriptions.* The system should support human-guided validation of algorithmically generated cluster descriptions, including visual comparison of cluster descriptions and individual sequences belonging to a cluster.

T8. *Human-driven refinement of clusters.* Hierarchical clustering algorithms construct a fixed dendrogram that defines the entire clustering hierarchy. This clustering process is based on computationally derived similarities. This inherently limits flexibility for a subsequent reconstruction of clusters. To overcome this limitation,

the system should support human-driven refinement of clusters based on event sequences' semantic similarities (e.g., descriptions derived by process mining).

- T9. *Responsive coordinated views and interactivity.* The visualization design should support coordinated views and incorporate rich interactions to facilitate in-depth inspection, comparison, and analysis with visual representations. When multiple coordinated views are presented, the layout should also be dynamically adjustable to accommodate shifting analyst focus and maintaining context.

Based on the identified design requirements and analytical tasks, we implement our proposed approach in a prototype VA system, reVISID. The architecture and key functionalities are detailed in Sec. 6.

5 ALGORITHMIC COMPONENTS OVERVIEW

Our overall pipeline is depicted in Fig. 2. We first outline the algorithmic components that comprise the preprocessing stage: extraction of interaction event sequences, similarity computation, hierarchical clustering, process model discovery, and conformance analysis.

5.1 Preprocessing of SiN Log

Each user interaction within SiN is recorded as a discrete event entry, timestamped, and stored in a JSON format. While interactions are logged at millisecond resolution, we simplified them to a second-based scale to reduce the cost of sequence similarity computation. As this simplification resulted in multiple identical events occurring within the same second, we further removed these consecutive identical events. Our primary analysis focus lies on how individuals navigate and interact with the exhibit interface, rather than examining the specific attributes being explored (e.g., selected gender or year). Drawing also on insights from our prior collaboration with domain experts [70], we extract four core event types from the logs, deemed essential for characterizing interaction strategies in the data. These are “Open” infopanel, “Revisit” infopanel, “Year Change” attribute, and “Gender Change” attribute.

5.2 Sequence Similarity

Following preprocessing, we use DTW to compute event sequence similarities. However, DTW is designed to operate on *continuous* time series data. Thus, our similarity computation first generates a continuous-time series-like structure for each event type. To do so, we insert intermediate events by duplicating the most recent event at one-second intervals until the subsequent event occurs. This temporal resolution is consistent with our preprocessing strategy, which aggregates the data at a one-second resolution, thereby avoiding excessive granularity at the millisecond scale. Given that the interaction log comprises four primary event types, as described in Sec. 5.1, this process results in four distinct univariate time series per user, each corresponding to one of the event types.

To compare each pair of users' multiple univariate time series with DTW, we can employ the dependent or independent variant of DTW [54]. The dependent DTW (DTW_D) assumes that all variables are mutually dependent, enforcing the same warping path across all variables. Due to this enforcement, DTW_D is less effective in scenarios where temporal lag or asynchronous behavior exists across event types [54]. In contrast, the independent DTW (DTW_I) computes a separate alignment path for each univariate time series, allowing separate warping paths for each variable. This is more appropriate when variables are lagged or loosely coupled. The SiN log dataset falls into such conditions since users cannot simultaneously engage with the interface and change attribute filters. Therefore, DTW_I is utilized.

Finally, both DTW_D and DTW_I fundamentally rely on the same underlying similarity computation. The dependent variant (DTW_D) operates analogously to traditional DTW for univariate time series, utilizing the cumulative squared Euclidean distance across the shared alignment path. Conversely, the independent variant (DTW_I) computes DTW alignments separately for each dimension and aggregates the resulting distances, each based on Euclidean distance [54]. However, in symbolic sequences (e.g., categorical values), such distance metrics become semantically inappropriate, as the magnitude of numerical encodings can influence the similarity assessment between the discrete

events. To address this, we implement a custom distance function that returns 0 for matches and 1 for mismatches, as follows:

$$\text{customDist}(e_1, e_2) = \begin{cases} 0, & \text{if } e_1 = e_2 \\ 1, & \text{if } e_1 \neq e_2 \end{cases} \quad (1)$$

After computing DTW distances, pairwise similarities are obtained across all sequences, resulting in a symmetric $n \times n$ distance matrix, where n is the number of event sequences/users. This matrix serves as the input for subsequent hierarchical clustering.

5.3 Hierarchical Clustering

We use hierarchical clustering to avoid predefining the number of clusters and enable flexible exploration across multiple levels of granularity. This is well-suited, as we lack prior knowledge about the uncontrolled, in-the-wild data. It also produces a dendrogram showing inter-cluster relationships (T5), which is commonly applied to time-series data [3]. A complete hierarchy is constructed by applying complete-linkage agglomerative hierarchical clustering [30]. We utilize a partial dendrogram to construct triangular glyph representations to enable flexible, stepwise exploration of the clustering structure (Sec. 6.1.1).

5.4 Cluster Description

After clustering, it is essential to generate a concise and interpretable summary of each cluster's interaction behaviors (T3). To this end, we leverage process mining techniques, specifically process maps [58]. A process map typically consists of nodes (activities) and directed arcs that capture precedence relations, indicating that one activity directly follows another. We adopt the directly follows graph (DFG) model as the basis for our process representation.

We use the Python library PM4Py [6] and its built-in functionality to derive a DFG from each cluster's event log. Note that, unlike the sequence similarity computation, we utilize the preprocessed event log data without applying temporal interpolation, preserving the original event states. The resulting process graph is then post-processed to extract nodes and links from the PM4Py output. We impose a fixed ordering of nodes within the process graphs to enhance visual comparison and identification of similarities and differences between process models (T6). This output is subsequently used for front-end rendering and interactive visualization, as described in Sec. 6.1.2.

5.5 Model Conformance

Conformance checking serves as a natural extension within the process mining paradigm, offering a principled method to assess the consistency between observed behaviors and the discovered models. To evaluate how well individual event sequences align with a representative process model, we perform conformance checking. This analysis supports the identification of potential outliers and the integration of domain expertise into the cluster interpretation process (T7).

Given that we utilize the output of DFGs as our process model representation, we adopt a conformance measure specifically designed to compare individual traces against DFGs. To this end, we leverage the alignment-based [1] conformance checking submodule of the PM4Py library [6], which applies the alignment algorithm tailored to DFGs to compare individual log/trace objects and a process model. Each sequence is assigned a fitness score, quantifying the degree of alignment with the model. This score can be interpreted as the degree of behavioral consistency between each sequence and the expected direct-follow relations in the model. These scores enable the ranking of sequences (T4), allowing analysts to quickly detect the most and least conforming sequences within a cluster, as described in further detail in Sec. 6.2.

6 SYSTEM OVERVIEW AND DESCRIPTION

The primary analytical pipeline facilitated by our system comprises four iterative stages, as illustrated in Fig. 2. The reVISID system comprises two primary components. The CLUSTER AND PROCESS VIEW corresponds to the second and third stages of the pipeline, supporting the exploration and interpretation of cluster structures and their associated process models. The SEQUENCE VIEW is primarily aligned

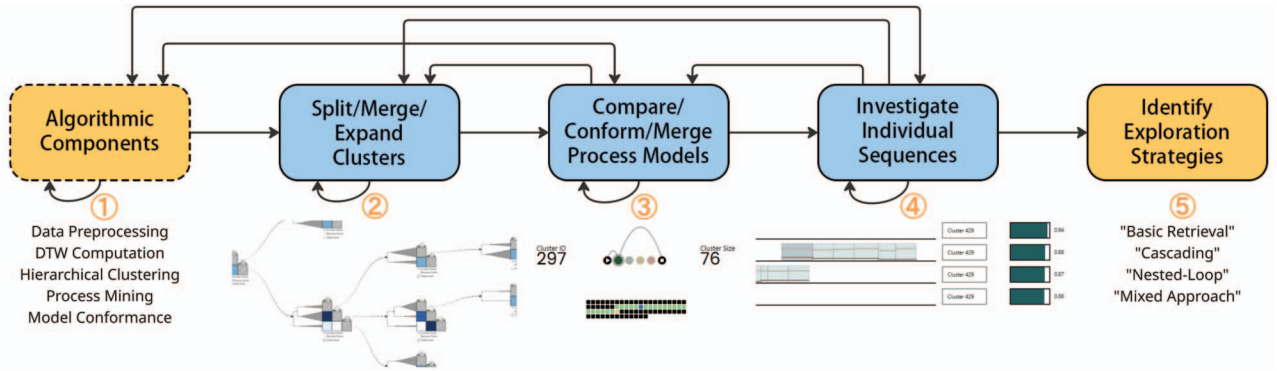


Fig. 2: Our analytical pipeline comprises five interconnected stages. (1) involves core algorithmic components, primarily executed in the backend, including data preprocessing, DTW-based similarity computation, hierarchical clustering, process model discovery through process mining, and conformance analysis. (2) Marks the beginning of the VA workflow, wherein users explore and refine the clustering hierarchy by performing operations such as splitting, merging, or expanding clusters. (3) Process models are computed for the clusters, which users can compare, validate through conformance checking, and merge based on domain knowledge. (4) enables detailed inspection of individual sequences. Finally, users can identify and characterize meaningful exploration strategies. The pipeline also supports iterative exploration and refinement across several stages.

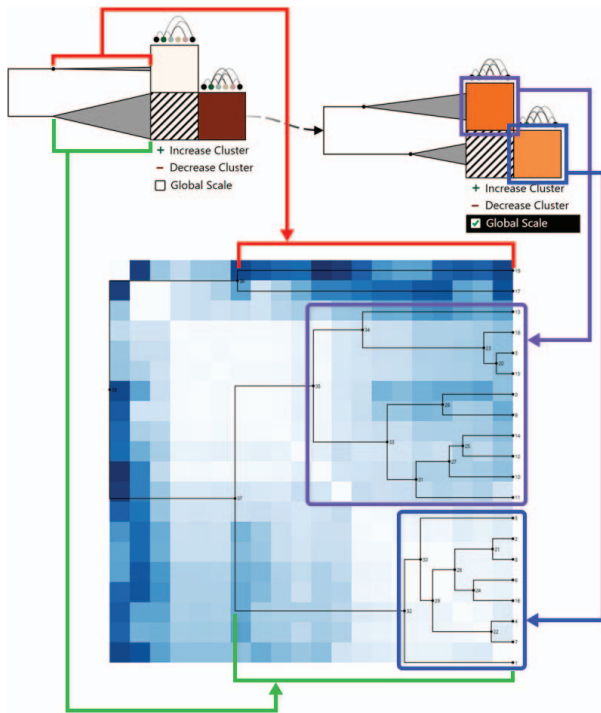


Fig. 3: Illustrative example using a small dataset to demonstrate how the traditional dendrogram structure is encoded using triangle glyphs, along with a breakdown of the visual components represented by each glyph.

with the third stage, facilitating detailed inspection of individual event sequences relative to the extracted models. Each view integrates a set of dedicated controls and a coordinated set of interactive visualizations, enabling various interactions such as visual customization, data filtering, and selecting data points of interest, to facilitate exploration of the uncontrolled, unstructured interaction log data within each view.

6.1 Cluster and Process View

The CLUSTER AND PROCESS VIEW, as illustrated in Fig. 1 (top), integrates multiple functionalities to support the aforementioned analysis tasks. This view comprises two primary components, a *Cluster Pane* and a *Process Description Pane*. The latter component is configured and controlled through the former. To elucidate its structure and functionality, we describe each pane and its corresponding controls.

6.1.1 Cluster Pane

The left pane (Fig. 1.1) presents a high-level summary of the top levels of the hierarchical clustering structure (T1). Previous work has explored aggregated dendrogram [40] and dendrogram-matrix representations [11]. Inspired by these, we introduce a summarizing *cluster matrix* representation encoded using a combination of a partial dendrogram with abstract isosceles triangular glyphs and a lower-triangular similarity matrix (see Fig. 3), enabling users to comprehend intra- and inter-cluster relationships and structural conciseness at a glance (T5).

In a traditional dendrogram, the branches reflect the original hierarchical relationships among individual data points or clusters, while the vertical height of each merge encodes the dissimilarity between two child clusters. We map the dendrogram hierarchy in the proposed cluster matrix representation in which the cluster nodes represent sub-clusters that are interactively expanded to reveal the next hierarchy level (T1). Intra-/inter- similarity is mapped on the color of each cluster matrix node, forming a lower-triangular similarity matrix that conveys pairwise cluster relationships. This design choice preserves screen space (T1) while maintaining the consistency of the matrix (T5). We utilize a partial dendrogram to convey structural information from the hierarchical clustering, preserving the connectivity and their associated heights, thus reducing visual clutter (T1). An isosceles triangle glyph is rendered adjacent to each cluster node. The base width of the triangle glyph encodes the number of sequences within the cluster, and the height corresponds to the distance between the clusters at the next step of the hierarchy, as illustrated in Fig. 3, providing a visual cue for assessing cluster similarity. This design conveys both the cluster size and its relative position within the hierarchical structure (T5). It serves as a visual indicator to support user-driven refinement of clustering granularity, providing cues as to which clusters to merge or subdivide when the number of clusters is manually decreased or increased, respectively. Detailed information about cluster composition, such as its size and the identifiers of the sequences composing it, can be accessed on hover (T2). The triangle glyphs are also interactive and serve as entry points for expanding individual clusters (cf. Sec. 6.3). The number of clusters is determined by counting the subclusters that remain unexpanded.

The initial number of clusters is determined using a method analogous to the elbow criterion (T3). The point of the largest increase in linkage distance (i.e., the most significant “gap”) is identified to suggest a suitable cut in the hierarchy structure. Nevertheless, the system provides interactive flexibility, allowing users to manually adjust the number of clusters and explore the impact of increasing or decreasing this number. However, the triangle glyph height does not convey the internal conciseness of individual clusters. To address this limitation, we compute intra-cluster conciseness scores by averaging all pairwise distances among sequences within each cluster, and inter-

cluster dissimilarity scores by averaging distances between sequences across clusters. Since these two measures capture fundamentally different aspects of cluster structure and are not directly comparable, we provide a toggle mechanism to enable selective display of intra- or inter-cluster scores through color encoding on the matrix (T5). Finally, to effectively use the remaining space of the matrix, we incorporate simplified/compressed process model thumbnails to provide a glimpse of the behavioral patterns within each cluster (T3).

6.1.2 Process Description Pane

The middle pane (Fig. 1.2) presents cluster descriptions derived by process mining (T3) and facilitates cluster description comparison (T6). These descriptions are visualized as arc diagrams and follow the same visual encoding as the process thumbnails used in the *Cluster Pane*.

A fixed number and fixed ordering of the nodes is enforced to enhance visual comparability across clusters (T6). Each process model consistently begins with a “Start” node, represented by a triangle symbolizing a play icon, and terminates at an “End” node, depicted as a square resembling a stop icon. All intermediate nodes are color-coded distinctly according to the four defined event types. Directed arcs between node pairs represent transitions within the process. The flow direction is encoded via spatial positioning; arcs positioned above the node axis indicate forward transitions from left to right, while arcs below signify transitions in the reverse direction. Arc thickness encodes the frequency of transitions between the nodes. To capture and visually integrate self-loop behaviors for each node, we utilize the border thickness, where thicker borders denote a higher number of self-transitions. Arcs within the process model visualization can be filtered through the controls provided in the *Cluster and Process Control Panel*, enabling users to focus on prominent transitions and reduce visual complexity.

To the left of each process model, the corresponding cluster ID is displayed, while the number of sequences in the cluster is shown to the right. This layout reinforces the association between each process description and its respective cluster (T9). Additionally, small squares rendered beneath each arc diagram provide a supplementary visual cue of cluster size. If a characterization of log sessions exists, it can be mapped to the color of these. These enhance visual discernibility and comparison across models. On the far right of each process model, a selection checkbox allows users to initiate model merging operations and link the cluster to detailed sequence-level visualizations (T9).

6.1.3 Process Merging Pane

The rightmost pane (Fig. 1.3) will dynamically display the process models that have been manually selected for merging based on expert-driven assessment (T8). These models are rendered using the same arc diagram encoding introduced in the *Process Description Pane*, ensuring visual consistency across views (T9). The models shown in this pane represent candidate clusters that, upon detailed examination and comparison, were identified by users as exhibiting sufficiently similar exploration patterns or strategies. These merged models can subsequently be exported, ready for downstream analysis or further synthesis. Further details regarding interactions and coordinated linking between panes and views are described in Sec. 6.3.

6.2 Sequence View

The *SEQUENCE VIEW*, as illustrated in Fig. 1 (bottom), is designed to address the analytical tasks centered on bridging the cluster process models and the underlying individual event sequences. This view comprises two primary components, a *Sequence Control Panel* and a *Sequence Visualization*. The latter component is configured and controlled through the former.

While process models represented in DFGs provide a structural abstraction of interaction flows within clusters, they inherently disregard precise temporal information. To mitigate this limitation, we employ a consistent color-coding scheme across the sequence and process model visualizations, enabling users to efficiently associate individual actions with their corresponding modeled behaviors. The color scheme is constructed using similar hues with variations in saturation, enabling users to easily distinguish the boundary through perceptual contrast. For

example, the “Open” and “Revisit” nodes in the DFG (dark and light green, respectively; see Fig. 1.3) correspond to identically color-coded bars in the sequence visualization (Fig. 1.4), facilitating direct visual association between modeled behaviors and individual sequence actions. To represent individual sequences, we adopt the nested juxtaposition design from our previous work [70], which enables the visualization of multiple event attributes in a temporally ordered layout (T4). This reveals specific actions performed within each sequence, thereby supporting the identification of representative, deviating, or outlier sequences. Domain experts are empowered to perform informed, manual validation of sequence-to-model conformance and assess whether particular subclusters or sequences are consistent with the process model (T7).

To support traceability and enhance interpretability, each individual sequence is annotated with its corresponding cluster identifier (T4). This annotation enables users to determine the sequence and respective cluster relationship quickly. The system supports coordinated highlighting to further assist in navigating and relating clusters to their constituent sequences (T9). When a cluster is hovered over in the *Process Description Pane*, all respective sequence cluster IDs associated with that cluster are highlighted. This interactive linkage facilitates comparison across views and supports identification of sequences relevant to a specific cluster or process model.

6.3 Interaction with reVISID

The interactions facilitated by the system correspond to stages 2 through 5 of the analytical pipeline Fig. 2. While the workflow generally follows a sequential progression, it is inherently iterative. Users may repeatedly adjust the number of clusters in stage 2 (e.g., by increasing or decreasing the dendrogram cut), proceed to stage 3 to compare the resulting process models, and examine individual sequences in stage 4. If the outcomes at any stage are deemed unsatisfactory, users can iterate and refine previous steps, enabling a flexible and exploratory analysis process. Finally, users arrive at a set of identified exploration strategies by selectively merging and exporting process models.

The *Cluster Pane* within the CLUSTER AND PROCESS VIEW supports stepwise exploration of the hierarchical clustering structure. Users can inspect individual clusters in detail, accessing information such as cluster size, constituent sequence identifiers, and intra-cluster conciseness (T2). Manual control over the number of clusters is provided by moving the cut level of the dendrogram upward or downward, effectively adjusting the granularity of the clustering (T9). The height of each triangle glyph serves as a visual indicator of the distance between two subclusters, offering cues as to which cluster is to be split or merged when adjusting the cluster count. When multiple clusters are visible, users can assess inter-cluster similarity and refer to the associated simplified process model thumbnails to evaluate whether structurally similar clusters can potentially be grouped (T5, T8). The detailed process model corresponding to each cluster is displayed in the *Process Description Pane*. Users can examine the process model in detail and compare whether some of the models are similar (T6). If a particular subcluster contains a large number of sequences, as indicated by a wide triangle glyph, and exhibits low conciseness with limited interpretability in the corresponding process model, it potentially requires further investigation. By default, the triangle glyph width is scaled locally concerning the current set of subclusters being examined. However, it could mislead users when performing cross-cluster comparisons due to inconsistent scale references. To mitigate this issue, users can toggle between local and global scaling modes, enabling more accurate interpretation of cluster sizes across varying hierarchical levels.

Users can expand a cluster by clicking on its associated triangle glyph. Upon expansion, child subclusters are shown with an estimated optimal number of subdivisions (T3). The process models in the *Process Description Pane* are updated accordingly. The linking between a specific cluster in the *Cluster Pane* and its corresponding process model in the *Process Description Pane* can be activated by hovering (T9).

To support detailed analysis of sequences within specific clusters, users can select the corresponding process models in the *Process Description Pane* and update the *SEQUENCE VIEW* to display the individual event sequences associated with the selected model, enabling users

to examine how the process unfolds over time (**T4**, **T9**). To further validate whether certain clusters can be grouped, users may activate the conformance checking functionality by selecting a specific model. This triggers an automatic evaluation of all sequences against the selected model and sorts them by descending conformance score, as illustrated in Fig. 1.5. Doing so allows users to assess how well each sequence aligns with the abstracted model (**T7**) and enables them, based on their domain expertise, to identify outliers or assess whether low-conforming sequences align semantically with the cluster or should undergo reclassification. This interactive and informed validation process reinforces the coherence of the cluster structure of the derived models.

The layout of the interface is designed to support flexible allocation of screen space across the three main panes (**T2**). Users can dynamically resize each pane via dragging interactions to prioritize specific analytical tasks. Within each pane, independent zooming and panning functionalities are provided, enabling seamless transitions between overview and detail [53] (**T2**). This interaction design facilitates scalable exploration. For example, users can zoom out in the *Cluster Pane* to gain a global perspective of the cluster hierarchy structure, including the positioning and hierarchical context of the currently selected cluster. Similarly, users can zoom out in the *Process Description Pane* to gain an overview of multiple process models, supporting easier visual comparison and selection.

This exploratory process is iteratively repeated until a sufficiently meaningful set of clusters with distinct and interpretable process models is identified. The selected process models can be merged and subsequently exported to the *Process Output Pane* for further analysis (**T8**). The visual representation of merged models adheres to the same arc diagram convention established in the *Process Description Pane*. The total number of constituent sequences resulting from the merge, as well as the identifiers of the originating clusters, are displayed.

7 CASE STUDIES

We conducted two workshops and analyzed the data with several domain experts, including the exhibit designer, developer, and our coauthors and visual learning and communication (VLC) experts, to demonstrate the applicability and usefulness of reVISID.

7.1 Interaction Log Datasets

We used two interaction log datasets in the use cases, including one unstructured, in-the-wild dataset and one semi-structured mixed dataset. **In-the-wild data.** We collected a wild dataset from the SiN exhibit throughout October 2024 to February 2025. The dataset comprises unstructured interaction logs collected in an uncontrolled manner from visitors as they freely engaged with the digital exhibit. As described in section 3, interaction data were collected only if visitors gave consent. A session was saved for every active interaction session after the exhibit went into idle mode, e.g., there had not been any interactions for the past five minutes. This means that each separate session, i.e., interaction log sequence, may include interactions from a single or multiple individuals engaging either consecutively or simultaneously. The final in-the-wild dataset used comprises 117 recorded sessions.

Mixed data. The second dataset we explored was a mixed dataset consisting of the newly collected in-the-wild data (117 sessions) and the structured dataset collected in our previous experimental study [70]. The structured dataset comprises interaction logs from 20 individuals, each solving five tasks involving increasingly complex cognitive learning processes. We have extracted each task from this dataset as a separate session, resulting in 100 interaction log sequences. The final mixed dataset thus comprises in total of 217 interaction sessions.

7.2 Case Study 1: Exhibit Design Analysis

Understanding how visitors interact with public exhibits is critical for optimizing system design and enhancing user experience. Equally important is adopting an educational perspective to interpret observed behaviors, uncover patterns, and formulate hypotheses about users' exploration and cognitive processes.

We conducted two collaborative workshops to analyze the collected in-the-wild interaction log dataset. The first workshop involved the

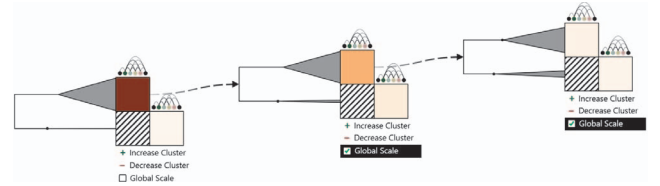


Fig. 4: A high-level summary of the hierarchical clustering structure for the in-the-wild dataset. At each level, two clusters are visualized, comprising a dominant (larger) and a smaller cluster representing outliers. The dominant cluster is iteratively expanded two levels, reaching a total depth of three. Exploration was concluded at this point, as no additional subclusters were deemed analytically meaningful for further investigation.

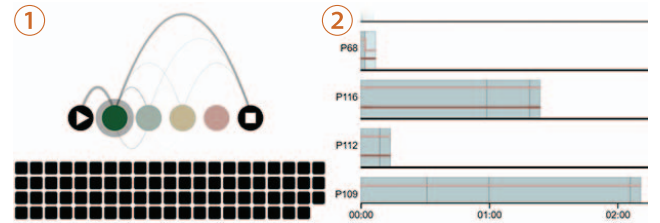


Fig. 5: The process model for the dominant cluster at the final hierarchical level reveals a pattern characterized by minimal user engagement. The majority of sequences within this cluster involve visitors opening one or more infopanel without performing subsequent actions in a short period of time. This behavioral pattern is visually encoded by (1) a prominent border around the “Open” node and a direct arc leading to the “End” node; and (2) brief interaction durations with no modifications to attributes.

original exhibit designer and one of the system developers, hereafter referred to as the “SiN design experts”. The second workshop was conducted with two co-authors and experts in visual learning and communication, hereafter referred to as the “VLC experts”. For clarity, we collectively refer to both groups as “experts”. During the workshops, we provided an overview of the analysis process enabled by the reVISID system, detailing the specific functionalities and interactions available to support exploratory data investigation. Following this, we commenced an exploration session where the authors were navigating reVISID under the instruction of the experts.

On startup, reVISID suggested splitting the dataset into two clusters, with a larger cluster consisting of 112 sequences and a smaller cluster consisting of 5 sequences, providing a starting point for analysis. The domain experts began their exploration by conducting a preliminary examination of the two initial cluster descriptions, by inspecting the process arc diagrams to gain an overview of the clustered interaction patterns. The larger cluster predominantly exhibited a pattern where users opened infopanel but did not perform further interactions within the exhibit, indicating minimal engagement. In contrast, the smaller cluster demonstrated more complex interaction behaviors, characterized by frequent attribute changes and repeated revisits to previously opened infopanel. Arc filtering was applied to refine the analysis further and differentiate more distinctly the prominent interaction patterns.

The experts then proceeded to the SEQUENCE VIEW to inspect the detailed interaction sequences composing each cluster, aiming to analyze variations in user interaction patterns with the exhibit. This investigation focuses on key aspects, including interaction behaviors (e.g., attribute changes), exploration patterns (e.g., the extent of infopanel engagement, frequency of revisits to previously opened infopanel), and interaction duration. The inspection revealed the smaller cluster was characterized by longer session durations (on average 15 minutes) with multiple user interactions with the exhibit. On the other hand, the larger cluster primarily comprised shorter sessions with very limited engagement, as indicated also by the cluster description. Upon viewing the difference in the interaction sequences of the smaller cluster, the SiN design experts commented on the possible presence of

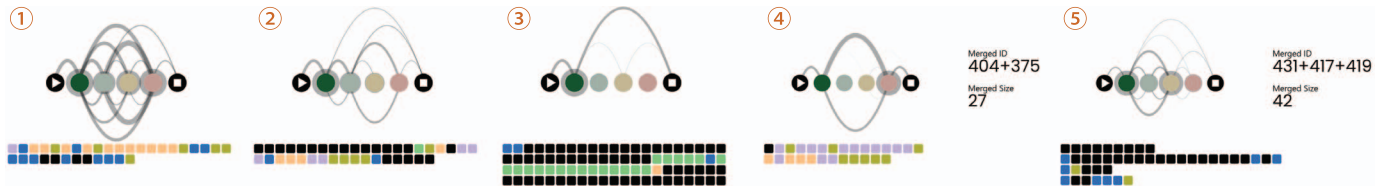


Fig. 6: Overview of the five clusters identified in the second case study. Detailed descriptions and analyses of each cluster are provided in Sec. 7.3.

on-site guides who regularly demonstrate exhibits to visiting groups, subsequently encouraging the visitors to engage and interact. These prolonged sessions, which may appear as outliers, could potentially be indicative of such interactions by guides and visiting groups.

Further partitioning of the larger cluster yielded two distinct subclusters. The dominant subcluster exhibited a prevalent interaction pattern where visitors primarily created new infopanel before disengaging from the exhibit, as illustrated in Fig. 5. In contrast, the smaller subcluster demonstrated similar prolonged engagement, characterized by frequent revisits to infopanel and changing of attributes. The analytical procedure was repeated for a third iteration, reaching three levels of depth before concluding, as illustrated in Fig. 4. Two key observations emerged from the analysis.

Key insight 1: Limited engagement with the exhibit. The exploration revealed that the most frequent interaction pattern consisted of users opening infopanel without engaging in subsequent actions or further exploration, as illustrated in Fig. 5. The SiN design experts concluded that the current version of the exhibit does not sufficiently capture or sustain user engagement beyond the initial interaction phase. They hypothesized that SiN’s functionality may be too complex for visitors to comprehend without explicit instruction, potentially explaining the smaller clusters characterized by longer interaction sessions, presumably corresponding to guided use. Similarly, the VLC experts noted that visitors appeared to struggle with sustained engagement, suggesting that the exhibit implicitly relies on a degree of self-directed or self-regulated learning competence, which may not be adequately supported by the current design. Visitors were unable to independently formulate tasks or questions to maintain engagement with SiN.

To address this limitation, the SiN design experts proposed the integration of an interactive onboarding guide to better support initial user engagement. The VLC experts further recommended the inclusion of structured guidance throughout the interaction experience. This includes embedding cognitively engaging tasks, reflective questions, or exploratory prompts that can be addressed through direct interaction with the exhibit. Prior work has demonstrated the value of such task-driven approaches and their resulting interaction patterns [70].

Key insight 2: Disparity in attribute utilization. The second critical observation concerned the significant discrepancy in the usage of different attributes in the infopanel. Specifically, the *Gender* attribute was consistently substantially underutilized compared to the *Year* attribute. The SiN design experts hypothesized two potential explanations for this behavior, both of which raise important concerns. One explanation could be that visitors fail to recognize that the *Gender* attribute, which is presented as small circular icons, is interactively adjustable. Whereas the *Year* attribute, presented as a traditional slider, is more immediately perceivable as adjustable. This hypothesis suggests a usability issue that can be directly addressed through interface design improvements. A second explanation is that the lack of interaction with the *Gender* attribute may indicate a lack of user interest in exploring gender-related data. This hypothesis was particularly significant given that the central objective and original motivation of the exhibit was to support the exploration of gender differences and raise awareness about gender equality in Sweden. This indicates underlying structural design limitations in the exhibit that do not have trivial or straightforward solutions.

Additionally, the VLC experts noted that visitors expected SiN to support access to implicit data information, such as trends, based on previous conversations [70]. Visitors expressed a desire to derive insights extending beyond the static representations provided by traditional

statistical tables and graphs. Specifically, they sought to address more personal questions that require integrating multiple data points and calculating trends. Therefore, the exhibit must support curiosity-driven inquiry, providing implicit information, addressing visitors’ personal needs and motivations, and delivering structured learning experiences.

7.3 Case Study 2: Validation of reVISID

As an extension to the second workshop with the VLC experts, we proceeded to explore the mixed dataset. Given that the in-the-wild data exhibited relatively simple and limited interaction patterns, this mixed dataset provided an opportunity to validate whether reVISID could effectively support the identification of more sophisticated exploration strategies, such as those previously uncovered in the structured dataset [70]. To facilitate validation and comparative analysis, we introduced color coding for session logs displayed in the *Process Description Pane*. Specifically, session logs from the structured dataset were color-mapped according to the five predefined tasks executed during the controlled experiment (Task 1 ■, Task 2 ■, Task 3 ■, Task 4 ■, and Task 5 ■), while those from the in-the-wild dataset were rendered in black by default. This visual distinction enabled clear differentiation between the datasets and supported the identification of known exploration strategies within the mixed clustering hierarchy.

We explored the mixed dataset using a similar progressive approach as in the previous analysis. Guided by visual indicators of cluster conciseness and size, we selectively expanded specific branches of the hierarchy, reaching a depth of at least three levels (see Fig. 1.1). The inclusion of both structured and unstructured data introduced greater variability in interaction patterns, which became evident during exploration. The resulting clusters exhibited clearer separation and suggested more distinct exploration strategies. This process further hinted that certain in-the-wild sequences may exhibit exploration strategies similar to those identified in the controlled experiment. Detailed inspection using the *SEQUENCE VIEW* validated this observation. By examining similarities in the cluster descriptions and utilizing the system’s merging functionality, we ultimately converged on a refined set of five clusters. These clusters reflected well-differentiated patterns of interaction behavior, as illustrated in Fig. 6, demonstrating reVISID’s capability to support identification of meaningful strategies within heterogeneous data. A detailed description of each cluster is presented below.

1. The first cluster demonstrates a highly active and diverse interaction pattern. The associated arc diagram reveals a densely connected structure, with transitions between nearly all nodes, resulting in a visually complex representation. This pattern reflects a combination of multiple exploration strategies, primarily characterized by both “cascading”-like [70] and comparison behaviors. In this cluster, most visitors either iteratively create infopanel, adjust attributes, and repeat the process, or open multiple infopanel in succession and subsequently revisit them to adjust the attributes of each. This interpretation is supported by a high concentration of Task 3 sequences (■), which previously demonstrated the “cascading” behavior.
2. The second cluster presents less activity in the interaction. No transitions are observed from the *Gender Change* back to the *Open* node, suggesting that users primarily open multiple infopanel, and subsequently revisit them to adjust the gender attribute. This behavior is aligned with the “nested loop” strategy identified in prior work, and is further corroborated by the high presence of Task 4 sequences (■) within this cluster.
3. The third cluster reflects a minimal interaction strategy, wherein

users predominantly initiate infopanel and disengage thereafter. This behavior is visually indicated by the pronounced self-loop on the *Open* node and a strong transition leading directly to the *End* node in the corresponding arc diagram. Such a pattern closely aligns with the prevailing behavior observed in the uncontrolled, in-the-wild dataset and resembles the interaction strategy identified in Task 1 (■) from our previous controlled experiment, where participants engaged in a basic inspection task.

4. This merged cluster reveals a comparatively simple interaction pattern characterized by the creation of individual infopanel, immediately followed by adjustments to the gender attribute, and subsequent transitions back to the *Open* node. According to our earlier definitions, this reflects a more streamlined variant of the “cascading” exploration strategy, albeit with fewer overall interactions. This strategy was prominently observed during Tasks 2 and 4 (■) in our prior controlled experiment, which is further confirmed by the distribution of session logs within this cluster.
5. The last merged cluster reveals a similar interaction pattern to Cluster 4, but with user engagement primarily focused on the year attribute. Due to the use of a slider interface for adjusting the year attribute, the resulting process model exhibits increased transitions to the *Revisit* node, leading to repeated revisit events. Interestingly this cluster has a high presence of task 5 (■) sessions, indicating that the change of year was prioritised for explorative tasks [70] involving more than simple comparison.

7.4 Reflections from a Learning Theory Perspective

The case studies provided the means to evaluate reVISID’s usefulness through a theoretical validity approach, using the learning theory lenses of Bloom’s taxonomy [7] and the three-factor situational interest model [39]. Bloom’s taxonomy classifies learning into six levels, progressing from basic to complex: remembering, understanding, applying, analyzing, evaluating, and creating. The three-factor situational interest (SI) model explains engagement through triggered interest, maintained feeling-related interest, and maintained value-related interest. Both theories are meaningful tools for evaluating and reflecting on the pedagogical design and implementation of interactive exhibits.

Findings from using reVISID suggest that most in-the-wild users demonstrated minimal engagement, indicative of lower-level cognitive processing per Bloom’s taxonomy. This underscores the need for scaffolding strategies to foster deeper interaction. In contrast, longer sessions with repeated attribute changes may reflect higher-order cognitive engagement. According to the SI model, sustained interaction is more likely when users find the content personally engaging [39]. The clustering and process mining features produce results that align with learning theory and are relevant to educational experts. By revealing interaction patterns informed by cognitive and motivational frameworks, reVISID supports evidence-based design improvements.

8 DISCUSSION

While our VA system supports exploration of both structured and unstructured interaction logs and meets the analytical goals, design considerations and limitations are relevant to discuss. This section highlights these and also considers potential future directions of research.

Expressiveness vs. effectiveness. In this work, we utilize arc diagrams with fixed node ordering to represent process models, facilitating compact visual comparisons across clusters. While this enhances readability and comparison by retaining consistency of representation across multiple process graphs, it reduces expressiveness [10] compared to formal representations, e.g., Petri nets. These alternatives offer richer semantics, such as parallelism and branching logic, but often produce complex, less interpretable graphs [44]. Given our goal of enabling interactive model comparison, we prioritized concise, consistent visual encoding to support exploration without overwhelming the user.

Generalizability of clustering techniques. In this work, we explored agglomerative hierarchical clustering, leveraging its tree-like structure to support stepwise navigation, refinement, and user-guided merging of clusters. However, partitional algorithms (e.g., k-means) operate via iterative assignment and centroid updates, producing flat clusterings

without inherent hierarchical structure. Consequently, they lack a linkage structure (e.g., dendrogram) necessary for supporting our visual navigation model. On the other hand, unlike agglomerative methods, divisive approaches do not require full precomputation of the hierarchy, which may offer improved scalability for large datasets. Top-down approaches, such as bisecting k-means [52], could be an alternative to experiment further. Since the choice of clustering algorithm may lead to different groupings, future work could include a comparative evaluation with alternative strategies, such as divisive hierarchical clustering (e.g., bisecting k-means), to assess differences in structure quality, computational efficiency, and usability. Additionally, interaction improvement can further facilitate exploration and interpretation of cluster hierarchies, aiding users in understanding which process models fall under a given cluster. One possibility would be to implement interactive features, such as hover-based brushing and linking, to reveal the process models associated with all descendant subclusters.

Generalizability to other domains. Our method has been developed and evaluated within the context of interaction logs. While the underlying framework is domain-agnostic, one future study could be to evaluate its generalizability to other interaction datasets, such as online learning platforms, where exploration strategies may differ significantly.

Alternative validation techniques. We currently employ conformance checking to assess alignment between individual sequences and representative process models. An alternative could be to use precision and fitness metrics with each model. Fitness values can indicate how well the model reflects the observed behavior, and precision can quantify the extent to which the model generalizes beyond the log data.

Follow-up evaluation. In addition to the case studies, further empirical validation is needed to assess the impact of our findings. A promising direction involves follow-up analysis after launching the interactive guide within the exhibit, comparing in-the-wild interaction patterns pre- and post-update. A key dataset limitation is the ambiguity of session boundaries, as sessions may involve multiple users interacting consecutively or simultaneously, introducing noise into strategy extraction. Future work could explore automated segmentation methods to disentangle such composite sessions and improve analytical accuracy.

Scalability. Potential scalability concerns in our system design arise both in terms of visual representation and memory usage when handling larger or more complex interaction data. The current DFG representation uses four high-level event types to balance interpretability and expressiveness, effectively capturing user exploration strategies in uncontrolled settings. A more detailed event taxonomy could support finer-grained analysis, but risks visual clutter and reduced clarity of representations. This work focuses on behavioral patterns rather than specific contextual parameters or attribute values. Additionally, system-level scalability is limited by memory and rendering constraints when visualizing large interaction datasets. To maintain responsiveness, we show only sequences relevant to a selected cluster and employ visual abstractions, such as partial dendrograms, heatmaps, and glyphs, to reduce complexity. While this improves performance, it limits the global overview, highlighting a broader trade-off between scalability and interpretability. Future work may address this through adaptive or level-of-detail techniques [68]. Finally, DTW is used to measure sequence similarity for its robustness to varying sequence length [48], future work could investigate normalization to improve robustness [36].

9 CONCLUSION

This paper has presented a VA approach that integrates hierarchical clustering and process mining to support the stepwise, interactive exploration of unstructured interaction log data. We employ a tailored DTW-based similarity measure and implement the proposed approach in the VA prototype system, reVISID. The system facilitates (1) incremental exploration and navigation in the cluster hierarchy; (2) interpretation and comparison of clusters and process models; (3) investigation and connection between the cluster and individual sequences; and (4) incorporation of domain knowledge to identify potential exploration strategies. The applicability and effectiveness of the proposed approach were demonstrated through two case studies involving the exhibit designer, developer, and domain experts.

SUPPLEMENTAL MATERIALS

All supplemental materials are available on OSF at <https://osf.io/6az29/>, released under a CC BY 4.0 license.

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